

WEEK 6 SKILL OSSP

Part 1 — Backslash Escaping / Special Character Processing

1. Created a new folder for week 6 skill

```
sainath@LAPTOP-GJ94VP0B:~$ mkdir week6_s
sainath@LAPTOP-GJ94VP0B:~$ cd ~/week6_s
sainath@LAPTOP-GJ94VP0B:~/week6_s$ pwd
/home/sainath/week6_s
```

2. Basic Backslash Escaping

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo "Hello\nWorld"
Hello\nWorld
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo -e "Hello\nWorld"
Hello
World
```

3. Escaped Spaces

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo Hello\ World
Hello World
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo Hello World
Hello World
```

4. Escape Special Characters

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo \*
*
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo \&
&
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo \|
|
```

5. Preserve Characters

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo '$HOME'
$HOME
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo "$HOME"
/home/sainath
```

6. Validate Parser Output

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ printf '<%s>\n' Hello\ World
<Hello World>
sainath@LAPTOP-GJ94VP0B:~/week6_s$ printf '<%s>\n' "Hello World"
<Hello World>
sainath@LAPTOP-GJ94VP0B:~/week6_s$ printf '<%s>\n' 'Hello World'
<Hello World>
```

1. Test Complex Input

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ echo "Hello\ World \$USER \& \*"
Hello\ World $USER \& \*
```

Summary:

The experiment demonstrates how backslash escaping is used in Linux shell processing to preserve spaces and special characters. Escape sequences such as `\n` were tested, along with escaped spaces, shell metacharacters, quoted strings, and complex inputs.

The outputs were validated using `printf` to observe how the shell parses individual arguments. The experiment confirms that escaping and quoting can control how the shell interprets special characters and preserves intended input.

PART 2 — Child Processes

1. Creating the Child Process Program, Compiling and Execute

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ nano child_process.c
sainath@LAPTOP-GJ94VP0B:~/week6_s$ gcc child_process.c -o child_process
sainath@LAPTOP-GJ94VP0B:~/week6_s$ ./child_process
Parent Process: PID = 34
Child Process: PID = 35
Waiting for child process...
Executing command: ls
total 36
-rwxr-xr-x 1 sainath sainath 16264 Sep 17 17:29 child_process
-rw-r--r-- 1 sainath sainath 640 Sep 17 17:29 child_process.c
Child process completed.
```

2. Passing Arguments to a Child Process

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ nano child_process.c
sainath@LAPTOP-GJ94VP0B:~/week6_s$ gcc child_process.c -o child_process
sainath@LAPTOP-GJ94VP0B:~/week6_s$ ./child_process
Parent Process: PID = 42
Child Process: PID = 43
Running: echo Hello OSSP Week 6
Hello OSSP Week 6
Child process completed.
```

3. Final Process Check

```
sainath@LAPTOP-GJ94VP0B:~/week6_s$ ps
PID TTY          TIME CMD
 11 tty1          00:00:00 bash
 52 tty1          00:00:00 ps
```

Summary:

The experiment demonstrated backslash escaping and special character processing in Ubuntu. We tested escaped spaces, special symbols, escape sequences, character preservation using quotes, and validated how the shell parses normal and complex inputs using `printf`.

We also implemented process creation and execution in C using `fork()`, `exec()`, and `wait()`. The child process executed commands with arguments, the parent process managed the child, and execution errors were handled using `perror()`.